SYSDESIGN REFERENCE

Rate Limiting Tradeoff Matrix

A decision-oriented comparison of alternatives and failure modes.

Protect capacity and enforce fair use by admitting requests against a defined budget.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / RATE-LIMITING

SYSDESIGN

Rate Limiting Tradeoff Matrix

CORE IDEA

Protect capacity and enforce fair use by admitting requests against a defined budget.

Reference architecture decision chain

01 Token Bucket Vs Sliding Window	02 Scope	03 Distributed Counters	04 Rejection Behavior

Decisions to defend

+ Token bucket vs sliding window
+ scope
+ distributed counters
+ rejection behavior
+ and headers.

Failure modes to test

+ Clock skew
+ hot identities
+ bypass paths
+ and limiter-store failure.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Rate Limiting Tradeoff Matrix

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

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